

Large-Spin CFTs on Plane Waves

Codex

ABSTRACT: We study local unitary CFTs in the large-spin plane-wave limits of Komargodski, Mischosia and Popov. Conformal transport fixes the scalar two-point function as a Mehler kernel, and a scalar three-point function determines a normalized finite-spin matrix element in any CFT and the corresponding hypergeometric pp-wave limit. Heisenberg Wigner–Eckart kinematics determines the orbit dependence of correlators; the reduced matrices remain CFT-specific OPE data. The boundary-light-speed limit of Kerr–AdS produces the pressures $\mathcal{F}_3 = (r_+^2 - 1)^2/(16\pi G_4 r_+)$ and $\mathcal{F}_4 = (r_+^2 - 1)^3/(16\pi G_5 r_+^2)$ for $T > T_c = 1/(2\pi)$. The pressures undergo second- and third-order large- N transitions, and their Legendre–Fenchel transforms determine the canonical entropy envelopes. Both pp waves admit temporal functions and have noncompact causal diamonds. Cylinder completion supplies a microcausal operator net with self-adjoint evolution and a nonnegative one-spin generator. Double-spin lower boundedness reduces to the all-representation bound $\Delta \geq \max(k, \bar{k})$, whose proof remains open. For a chosen superconformal algebra and R -charge ray, the zero-grade roots determine the preserved superalgebra and the large-spin scaling of protected indices. Neutral Kerr–AdS₄ and Kerr–AdS₅ limits realize the two pp-wave boundary metrics. Their Fefferman–Graham coefficients fix the holographic stress tensors, and their horizon generators determine the thermal quotients.

KEYWORDS: Conformal Field Theory, Large Spin, Plane Waves, Heisenberg Symmetry, Kerr–AdS, Supersymmetric Index

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1 Large-spin limits and universal data

Let a local unitary CFT in dimension greater than two be placed on the Lorentzian cylinder. In three spacetime dimensions define $\tau = \Delta - J_z$ and take

$$J_z \longrightarrow \infty, \quad \frac{\tau}{\sqrt{J_z}} \text{ fixed.}$$

In four dimensions define $\tau = \Delta - J_1 - J_2$ and take

$$J_1, J_2 \longrightarrow \infty, \quad \frac{J_1}{J_2} \quad \text{and} \quad \frac{\tau}{(J_1 J_2)^{1/3}} \quad \text{fixed.}$$

Komargodski, Miscioscia and Popov (KMP) showed that these limits are described by plane waves and emergent Heisenberg symmetries [1, 2]. They derived the universal extensive thermodynamics and left five calculations open: correlators, thermal nonanalyticity, causality and positive energy, supersymmetry and indices, and large- N gravity with pp-wave boundary conditions.

Heisenberg symmetry determines kinematics, not OPE coefficients. A general CFT need not be supersymmetric or belong to a large- N family. Four-dimensional double-spin positivity requires the all-representation chiral bound of ref. [5], which is stronger than ordinary conformal unitarity for sufficiently chiral representations.

1.1 Results

Theorem 1.1 (Plane-wave results). *With the pp-wave theory defined by the conformal-cylinder scaling limit:*

1. *Conformal transport maps a scalar-primary vacuum two-point function to the pp wave. At fixed Heisenberg central momentum it is the Mehler kernel (2.10). For scalar primaries A, Φ , the normalized matrix element $\langle J; A | \Phi | J; A \rangle$ is the finite sum (2.17). Its local pp-wave limit is the hypergeometric function (2.18). Conformal three-point kinematics makes the derivation valid in interacting CFTs.*
2. *The Heisenberg algebra fixes selection rules, group-orbit dependence, and the recursions (2.11). Generic large- τ heavy-state correlators also contain reduced matrices on a multiplicity space. Those matrices are CFT-specific OPE data, not fixed by symmetry.*
3. *At large N , Kerr-AdS gives nonanalytic pressures (3.3) at $T_c = 1/(2\pi)$. Their Legendre-Fenchel transforms give the entropy functions (3.8)–(3.10).*
4. *The plane waves are stably causal and lack global hyperbolicity. The inherited conformal net is microcausal and its time generator is self-adjoint. The one-spin Hamiltonian is nonnegative. Lower boundedness of the four-dimensional double-spin Hamiltonian is equivalent, under large-spin additivity, to $\Delta \geq \max(k, \bar{k})$ for every local primary. The general bound is an open conjecture.*
5. *For a specified superconformal algebra and R -charge ray, the preserved superalgebra is the zero-grade centralizer (5.3). The protected three- and four-dimensional rays are given in (5.9) and (5.18).*
6. *The metrics (6.3) and (6.21) are exact Einstein solutions whose conformal boundaries are the KMP pp waves. Their largest roots are regular horizons. The asymptotic coefficients determine the holographic stress tensors.*

1.2 Conventions

The cross term $dt dx$ means $g_{tx} = 1/2$. We set the AdS radius to one and restore it in dimensionful formulas. We subtract the thermal-AdS classical action when quoting the leading holographic pressure. Lorentz representations in four dimensions are denoted by the numbers $(k, \bar{k})$ of undotted and dotted spinor indices.

2 Heisenberg kinematics and correlators

We transport vacuum correlators from the cylinder to the pp wave. Heisenberg symmetry determines the Mehler kernel and the orbit dependence of local matrix elements, but generic heavy-state correlators still require reduced OPE matrices.

2.1 Contracted geometry and normalization

The three-dimensional metric in the physical large-spin frame is

$$ds_3^2 = -(1 + y^2) dt^2 + dt dx + dy^2. \quad (2.1)$$

Removing the constant term by $x_{\text{un}} = x - t$ gives $ds_0^2 = -y^2 dt^2 + dt dx_{\text{un}} + dy^2$. Its Killing vectors

$$m = \partial_x, \quad p = \cos t \partial_y + 2y \sin t \partial_x, \quad q = \sin t \partial_y - 2y \cos t \partial_x \quad (2.2)$$

have x -components normalized by a factor of two, matching the oscillator pair $K_2 \pm iK_3$ of ref. [1]. For the Hermitian generators $M = -im$, $P = -ip$, and $X = -iq$,

$$[P, X] = 2iM, \quad [H, P] = -iX, \quad [H, X] = iP. \quad (2.3)$$

At $M = p_0 > 0$,

$$a = \frac{P + iX}{2\sqrt{p_0}}, \quad [a, a^\dagger] = 1, \quad [H, a] = -a. \quad (2.4)$$

The four-dimensional double-spin frame is

$$ds_4^2 = -dt^2 - 2dt dv - 4u dt dy + du^2 + dy^2. \quad (2.5)$$

Its Killing fields generate

$$\begin{aligned} V = \partial_v, \quad Y = \partial_y, \quad U = \partial_u - 2y\partial_v, \quad [U, Y] = 2V, \\ [H, U] = [H, Y] = [H, V] = 0. \end{aligned} \quad (2.6)$$

The full $\mathbb{H}_3$ commutes with the double-spin thermal Hamiltonian, so thermal Ward identities in that ensemble use the entire Heisenberg algebra. In the one-spin ensemble they use only the centralizer of H .

2.2 Scalar two-point kernel

The unshifted oscillator plane wave in any number of transverse directions,

$$ds_0^2 = -|z|^2 dt^2 + dt dx + dz^i dz^i, \quad (2.7)$$

is conformally flat. On a strip with $\cos t \neq 0$, the transformation

$$u = \tan t, \quad v = |z|^2 \tan t - x, \quad Y^i = \frac{z^i}{\cos t} \quad (2.8)$$

obeys $ds_0^2 = \cos^2 t (-du dv + dY^i dY^i)$. Weyl covariance therefore determines every scalar-primary two-point function from the flat interval.

Proposition 2.1 (Vacuum Mehler kernel). *Let $\mathcal{O}$ be a scalar primary of dimension Δ , normalized by $\langle \mathcal{O}(X) \mathcal{O}(0) \rangle = C_{\mathcal{O}} / (X^2)_{\text{W}}^{\Delta}$. On the oscillator wave,*

$$\langle \mathcal{O}(1) \mathcal{O}(2) \rangle_{\text{pw}} = \frac{C_{\mathcal{O}}}{[\mathcal{D}_{12}]_{\text{W}}^{\Delta}}, \quad \mathcal{D}_{12} = (|z_1|^2 + |z_2|^2) \cos T - 2z_1 \cdot z_2 + x_{12} \sin T, \quad (2.9)$$

where $T = t_1 - t_2$. With $\tilde{G}_p = \int_{\mathbb{R}} dx_{12} e^{ipx_{12}} G$, its branch on $0 < T < \pi$ is

$$\boxed{\tilde{G}_p(T; z_1, z_2) = \frac{2\pi C_{\mathcal{O}} e^{i\pi\Delta/2}}{\Gamma(\Delta)} \frac{\Theta(p \sin T) |p|^{\Delta-1}}{|\sin T|^{\Delta}} \exp \left[-ip \frac{(|z_1|^2 + |z_2|^2) \cos T - 2z_1 \cdot z_2}{\sin T} \right]}. \quad (2.10)$$

In the physical frame (2.1), the Fourier kernel is multiplied by e^{ipT} .

Proof. We insert (2.8) into the flat Wightman interval. The two primary Weyl factors cancel its powers of $\cos t_1 \cos t_2$, leaving (2.9). The distributional identity

$$\int_{\mathbb{R}} dx e^{ipx} (A + x \sin T - i0_{\text{W}})^{-\Delta} = \frac{2\pi e^{i\pi\Delta/2}}{\Gamma(\Delta)} \frac{\Theta(p \sin T) |p|^{\Delta-1}}{|\sin T|^{\Delta}} e^{-ipA/\sin T}$$

then gives (2.10). The shift $x_{\text{un}} = x - t$ gives the last phase. $\square$

The Wightman prescription transports the branch through the caustics $T \in \pi\mathbb{Z}$ and supplies the Maslov phases on adjacent strips. Fourier transformation of both insertions produces $2\pi\delta(p_1 + p_2)$. Formula (2.10) is the Stone–von Neumann Mehler kernel for the vacuum state. A thermal heavy-state correlator requires a density matrix or equivalent OPE data.

2.3 Ward identities and reduced matrices

Set $\Pi_i = P_i/\sqrt{2}$ and $\Xi_i = X_i/\sqrt{2}$, so that $[\Pi_i, \Xi_j] = i\delta_{ij}M$. Away from coincident insertions, the Heisenberg Ward identities are

$$\sum_{a=1}^n \mathcal{L}_M^{(a)} G_n = 0, \quad \sum_{a=1}^n \mathcal{L}_{\Pi_i}^{(a)} G_n = 0, \quad \sum_{a=1}^n \mathcal{L}_{\Xi_i}^{(a)} G_n = 0. \quad (2.11)$$

Spin matrices are included in $\mathcal{L}$ for spinning fields and local contact terms are restored at coincidences. The first identity becomes $\sum_a p_a = 0$ after Fourier transformation.

In the direct-integral decomposition for the noncompact central translation, the spectral fiber at nonzero central character $p > 0$ (whose vectors are delta-normalized generalized p -eigenstates) has the Stone–von Neumann form

$$\mathcal{H}_p = \mathcal{F}_p \otimes \mathcal{K}_p, \quad (2.12)$$

where $\mathcal{F}_p$ is the unique oscillator module and $\mathcal{K}_p$ is a CFT-dependent multiplicity space. If $\mathcal{O}(g) = U(g)\mathcal{O}(e)U(g)^{-1}$, then

$$\begin{aligned} \langle p', n', \alpha' | \mathcal{O}(g) | p, n, \alpha \rangle \\ = \sum_{r', r} D_{n'r'}^{(p')}(g) C_{r'\alpha', r\alpha}^{\mathcal{O}} D_{nr}^{(p)}(g)^*. \end{aligned} \quad (2.13)$$

Here $C^{\mathcal{O}}$ is the base-point reduced matrix. The Fourier mode $\mathcal{O}_q = \int dx e^{-iqx} \mathcal{O}(x)$ carries $q = p' - p$. A point-local insertion therefore superposes all q . Inserting complete sets in (2.13) produces products of reduced matrices for multi-operator correlators.

The group displacement has the matrix element

$$\langle n | e^{\zeta a^\dagger - \bar{\zeta} a} | n \rangle = e^{-|\zeta|^2/2} L_n(|\zeta|^2). \quad (2.14)$$

This Laguerre factor enters the representation matrix $D^{(p)}$ in (2.13). The CFT-dependent data sit in $C^{\mathcal{O}}$.

2.4 An interacting-CFT large-spin form factor

Scalar three-point kinematics determines a local matrix element in any CFT. Let A, Φ be unit-normalized real scalar primaries of dimensions Δ, δ , with

$$\langle A(x_1) \Phi(x_2) A(x_3) \rangle = \frac{C_{A\Phi A}}{|x_{12}|^\delta |x_{23}|^\delta |x_{13}|^{2\Delta-\delta}}. \quad (2.15)$$

For $z = e_1 + ie_2$, define the normalized state

$$|J; A\rangle = \frac{(z \cdot P)^J |A\rangle}{\sqrt{4^J J! (\Delta)_J}}. \quad (2.16)$$

At a point n of the unit sphere, let s^2 be the squared projection onto the $e_1 e_2$ -plane.

Proposition 2.2 (Exact local form factor). *For every nonnegative integer J ,*

$$\boxed{\langle J; A | \Phi_{\text{cyl}}(n) | J; A \rangle = C_{A\Phi A} \frac{J!}{(\Delta)_J} \sum_{\ell=0}^J \frac{(\Delta - \delta/2)_{J-\ell}}{(J-\ell)!} \frac{(\delta/2)_\ell^2}{(\ell!)^2} s^{2\ell}}. \quad (2.17)$$

Put $q = \delta/2$, $\lambda = \Delta - q$, $p = \epsilon J$, and $\theta = \pi/2 + \sqrt{2\epsilon} y$. If $q > 1/2$ and $\lambda > 0$, then

$$\begin{aligned} \mathcal{F}_\Phi(p, y) &:= \lim_{\epsilon \rightarrow 0} \frac{2\pi}{\epsilon} (2\epsilon)^q \langle J; A | \Phi_{\text{cyl}}(\theta) | J; A \rangle \\ &= \frac{2^{q+1} \pi C_{A\Phi A} p^{q-1} \Gamma(\Delta) \Gamma(2q-1)}{\Gamma(q)^2 \Gamma(\Delta + q - 1)} {}_1F_1(2q-1; \Delta + q - 1; -2py^2). \end{aligned} \quad (2.18)$$

Proof. Radial quantization and (2.15) give the generating function

$$\langle A | e^{bK_z} \Phi(n) e^{aP_z} | A \rangle = C_{A\Phi A} (1 - 2bse^{-i\varphi})^{-q} (1 - 2ase^{i\varphi})^{-q} (1 - 4ab)^{-\lambda}. \quad (2.19)$$

Taking the coefficient of $a^J b^J$ and dividing by the norm in (2.16) proves (2.17). The Euler integral for the resulting hypergeometric polynomial, followed by $J = p/\epsilon$ and $s^2 = \sin^2 \theta = 1 - 2\epsilon y^2 + O(\epsilon^2)$, is dominated for the stated range and gives (2.18). $\square$

Taking $A = \Phi = \varepsilon$ in the three-dimensional Ising CFT gives a nonconstant transverse profile determined by the interacting data Δ_ε and $C_{\varepsilon\varepsilon\varepsilon}$ [6]. The state in (2.16) has fixed $\tau = \Delta$, placing it at the $x = \tau/\sqrt{J} = 0$ endpoint of the required scaling surface.

On the thermodynamic ray, the cylinder-to-plane-wave normalization is

$$\phi = \epsilon x, \quad p = \epsilon J, \quad \mathcal{O}_{\text{pw}} = (2\epsilon)^{\Delta_{\mathcal{O}}/2} \mathcal{O}_{\text{cyl}}, \quad |p\rangle_\delta = \sqrt{2\pi/\epsilon} U_\epsilon |J\rangle. \quad (2.20)$$

The thermodynamic saddle has $J \sim \epsilon^{-2}$, so $p \sim \epsilon^{-1}$ and $\tau \sim \sqrt{J}$. In that regime the correlators require the large- p asymptotics of $C^{\mathcal{O}}$ in (2.13). If $\dim \mathcal{K}_p \geq 2$, a single Heisenberg representation admits multiple Hermitian base-point matrices, each giving distinct heavy-state correlators. Generic matrix elements $\langle J, \tau | \mathcal{O}_1 \cdots \mathcal{O}_n | J, \tau \rangle$ depend on the corresponding large- p OPE matrices.

3 Thermodynamic nonanalyticity

Finite angular regulators keep $\log Z$ analytic on every zero-free domain. Nonanalyticity appears only after the effective-volume limit $\epsilon_i \rightarrow 0$, and at large N the Kerr-AdS saddles produce second- and third-order transitions in the pressures.

3.1 Finite regulators and limiting singularities

For fixed positive angular regulators, define

$$\begin{aligned} Z_3(\beta, \epsilon) &= \text{Tr} \exp[-\beta(\Delta - (1 - \epsilon)J)], \\ Z_4(\beta, \epsilon_1, \epsilon_2) &= \text{Tr} \exp[-\beta(\Delta - (1 - \epsilon_1)J_1 - (1 - \epsilon_2)J_2)]. \end{aligned} \quad (3.1)$$

Whenever the regulated operator has a trace-class heat kernel, $Z(\beta)$ is holomorphic for $\text{Re } \beta > 0$, and $\log Z$ is analytic away from its zeros. The effective-volume limit $\epsilon_i \rightarrow 0$ permits zeros to condense. Large N can destroy local uniformity by the same mechanism.

The KMP normalization defines

$$\log Z_3 = \frac{2\pi\beta}{\epsilon} \mathcal{F}_3(\beta) + O(1), \quad \log Z_4 = \frac{\pi^2\beta}{2\epsilon_1\epsilon_2} \mathcal{F}_4(\beta) + O(\epsilon_i^0). \quad (3.2)$$

Symmetry fixes the extensive powers of ϵ_i . The functions $\mathcal{F}_d$ retain the spectrum and interactions of the CFT.

3.2 Kerr–AdS pressures and transition order

Let $r = r_+$ and set the AdS radius to one. Taking the angular velocities of Kerr–AdS to the speed of light at fixed r gives

$$\boxed{\begin{aligned} T_3(r) &= \frac{3r^2 - 1}{4\pi r}, & \mathcal{F}_3^{\text{BH}}(r) &= \frac{(r^2 - 1)^2}{16\pi G_4 r}, \\ T_4(r) &= \frac{2r^2 - 1}{2\pi r}, & \mathcal{F}_4^{\text{BH}}(r) &= \frac{(r^2 - 1)^3}{16\pi G_5 r^2}. \end{aligned}} \quad (3.3)$$

We apply the one-spin Kerr–AdS₄ limit and the double-spin Kerr–AdS₅ limit to the grand potentials

$$\begin{aligned} \mathcal{G}_4 &= \frac{(r^2 + a^2)(1 - r^2)}{4G_4 r(1 - a^2)}, \\ \mathcal{G}_5 &= \frac{\pi(r^2 + a_1^2)(r^2 + a_2^2)(1 - r^2)}{8G_5 r^2(1 - a_1^2)(1 - a_2^2)} \end{aligned} \quad (3.4)$$

using respectively $\mathcal{F}_3 = -\lim_{\epsilon \rightarrow 0} \epsilon \mathcal{G}_4 / (2\pi)$ and $\mathcal{F}_4 = -\lim_{\epsilon_i \rightarrow 0} 2\epsilon_1 \epsilon_2 \mathcal{G}_5 / \pi^2$. The Kerr thermodynamics and its Hawking–Page crossing are standard [7–9].

At leading order in $1/G$, background subtraction sets the thermal-AdS pressure to zero. The dominant branches are

$$\mathcal{F}_d(T) = \begin{cases} 0, & T \leq T_c, \\ \mathcal{F}_d^{\text{BH}}(r(T)), & T \geq T_c, \end{cases} \quad T_c = \frac{1}{2\pi}. \quad (3.5)$$

Writing $t = T - T_c$ and expanding at $r = 1$ gives

$$\mathcal{F}_3(T) = \frac{\pi}{4G_4} t_+^2 + O(t_+^3), \quad \mathcal{F}_4(T) = \frac{4\pi^2}{27G_5} t_+^3 + O(t_+^4). \quad (3.6)$$

The first discontinuous temperature derivative of $\mathcal{F}_3$ has order two, and that of $\mathcal{F}_4$ has order three, so the transitions are second- and third-order in the Ehrenfest sense. At finite N , the $O(N^0)$ gas branch and finite-volume rounding depend on the spectrum.

3.3 Large-spin entropy functions

We write the large-spin entropy functions as

$$\begin{aligned} \log \rho_3(J, \tau) &\sim \sqrt{J} G_3(x), & x &= \frac{\tau}{\sqrt{J}}, \\ \log \rho_4(J_1, J_2, \tau) &\sim (J_1 J_2)^{1/3} G_4(x), & x &= \frac{\tau}{(J_1 J_2)^{1/3}}. \end{aligned} \quad (3.7)$$

Eliminating the angular regulators at the inverse-Laplace saddle gives the universal Legendre–Fenchel relations

$$\boxed{\begin{aligned} G_3(x) &= \inf_{\beta > 0} \beta \left[x + 2\sqrt{2\pi \mathcal{F}_3(\beta)} \right], \\ G_4(x) &= \inf_{\beta > 0} \beta \left[x + 3 \left(\frac{\pi^2 \mathcal{F}_4(\beta)}{2} \right)^{1/3} \right]. \end{aligned}} \quad (3.8)$$

For the classical Kerr branches we set

$$A = \frac{1}{16\pi G_4}, \quad d = \left(\frac{\pi}{32G_5} \right)^{1/3}. \quad (3.9)$$

The black-hole parts of the transforms are

$$\boxed{\begin{aligned} x_3(r) &= \sqrt{2\pi A} \frac{r^2 + 1}{\sqrt{r}}, & G_3(r) &= 4\pi\sqrt{2\pi A} r, \\ x_4(r) &= d \frac{r^2 + 1}{r^{2/3}}, & G_4(r) &= 4\pi d r^{1/3}, \end{aligned} \quad r \geq 1.} \quad (3.10)$$

At the Hawking–Page endpoint,

$$x_{3c} = 2\sqrt{2\pi A}, \quad x_{4c} = 2d. \quad (3.11)$$

The classical Legendre–Fenchel envelope is $G_i(x) = 2\pi x$ for $0 \leq x \leq x_{ic}$, followed by the appropriate parametric branch in (3.10). The straight piece is the canonical concave envelope selected by $\beta_c = 2\pi$. Equality to a literal microcanonical coexistence entropy requires a matching large-deviation lower bound or an explicit phase-mixture construction. On the black-hole branches an independent area-law computation gives

$$\frac{d\mathcal{F}_3}{dT} = \frac{r^2 - 1}{4G_4}, \quad \frac{d\mathcal{F}_4}{dT} = \frac{(r^2 - 1)^2}{4G_5 r},$$

which agree with the regulated horizon entropy densities. Quantum gas states correct this classical envelope at order N^0 .

4 Causality and the pp-wave Hamiltonian

The pp-wave geometries are stably causal but not globally hyperbolic. Cylinder completion still yields a microcausal net with self-adjoint time evolution. The one-spin Hamiltonian is nonnegative, while double-spin positivity is equivalent to the chiral bound.

4.1 Temporal function and noncompact causal diamonds

The coordinate change

$$u = t, \quad v = -\frac{x}{2}, \quad \widehat{v} = v + \frac{u}{2}, \quad (4.1)$$

puts the one-spin metric (2.1) in the standard quadratic plane-wave form

$$ds^2 = -2du d\widehat{v} - |z|^2 du^2 + dz^i dz^i. \quad (4.2)$$

In the double-spin metric (2.5), denote the transverse KMP coordinate by w . The isometry to (4.2) is

$$z_1 = w \cos t + y \sin t, \quad z_2 = -w \sin t + y \cos t, \quad x = -2(v + wy). \quad (4.3)$$

The last normalization corrects the corresponding printed coordinate formula in ref. [1].

Proposition 4.1 (Causal classification). *The pp waves (2.1) and (2.5) are stably causal and have noncompact causal diamonds.*

Proof. On (4.2), the global function

$$\mathcal{T} = u + \arctan \frac{2\widehat{v}}{1 + |z|^2} \quad (4.4)$$

satisfies

$$g^{-1}(d\mathcal{T}, d\mathcal{T}) = -\frac{4}{(1 + |z|^2)^2 + 4\widehat{v}^2} < 0. \quad (4.5)$$

The strict negativity makes $\mathcal{T}$ a temporal function.

Take $p = (0, 0, 0)$ and $q = (\pi, 0, 0)$ in $(u, \widehat{v}, z)$. For every $R > 0$,

$$z_R(u) = R \sin u e_1, \quad \widehat{v}_R(u) = \frac{R^2}{4} \sin 2u, \quad 0 \leq u \leq \pi \quad (4.6)$$

is a future null curve from p to q , since $-2\widehat{v}'_R - |z_R|^2 + |z'_R|^2 = 0$. At $u = \pi/2$ it reaches $z = Re_1$. The unbounded parameter R makes $J^+(p) \cap J^-(q)$ noncompact and rules out global hyperbolicity. This null family realizes the critical quadratic case in the plane-wave classification [3]. $\square$

We define the global pp-wave theory by conformal pullback of the cylinder CFT. That pullback selects the causal structure of the pp wave, including the noncompact diamonds.

4.2 Locality and self-adjoint evolution for an interacting CFT

Let $\mathcal{A}_{\text{cyl}}(O)$ be the local operator algebra of the cylinder CFT. On every pp-wave conformal chart χ , define

$$\mathcal{A}_{\text{pw}}(O) = U(\chi) \mathcal{A}_{\text{cyl}}(\chi O) U(\chi)^{-1}, \quad (4.7)$$

with the appropriate Weyl action on fields and agreement on chart overlaps. Conformal maps preserve spacelike separation, so

$$[\mathcal{A}_{\text{pw}}(O_1), \mathcal{A}_{\text{pw}}(O_2)] = 0 \quad (O_1 \perp O_2). \quad (4.8)$$

The construction uses only conformal covariance of the cylinder net, so it applies to every local unitary CFT. Even-dimensional Weyl anomalies add local c-number contact terms and leave separated commutators unchanged.

The pp-wave time isometry belongs to the global conformal group, whose CFT representation gives a strongly continuous unitary group. Stone's theorem supplies its self-adjoint generator. Smooth Gårding vectors form a common invariant dense domain for the conformal Lie algebra. Nelson's analytic-vector theorem proves essential self-adjointness there for the displayed real linear combinations [14–16].

4.3 Positive energy and the chiral bound

For one large spin the time generator can be written in a light-cone conformal basis as

$$H_1 = P_u + P_v - K_v. \quad (4.9)$$

The conformal-algebra element $P_u - K_v$ is unitarily conjugate to the ordinary one-spin twist generator $D + 2M_{uv}$. Positive-energy representations have $P_v \geq 0$, which gives

$$H_1 \geq 0. \quad (4.10)$$

The double-spin generator is

$$H_2 = H_1 - L_{12}. \quad (4.11)$$

For a four-dimensional primary in Lorentz representation $(k, \bar{k})$, maximizing the KMP-oriented Cartan gives $\Delta \geq k$. The Hermitian-conjugate primary lies in $(\bar{k}, k)$ and supplies the orientation-reversed inequality. Both orientations require

$$\Delta \geq \max(k, \bar{k}). \quad (4.12)$$

Equation (4.12) is the open all-representation chiral-dimension conjecture of ref. [5], since ordinary conformal unitarity gives a weaker bound for sufficiently chiral representations and existing averaged-null-energy methods cover several families [4, 5].

Proposition 4.2 (Equivalence of the remaining energy problem). *Assume the standard large-spin additivity/cluster construction for multi-twist operators. Equation (4.12) makes the double-spin pp Hamiltonian bounded below. Any primary violating that bound generates a spectrum unbounded below.*

Proof. Under (4.12), every primary has nonnegative double-spin grade, and conformal raising operators have nonnegative grade, so the state space is lower bounded. A primary of grade $\tau_0 < 0$ produces large-spin multi-twist states whose grade is $n\tau_0 + o(n)$ by large-spin additivity [2]. Sending $n \rightarrow \infty$ makes the spectrum unbounded below. $\square$

Lower boundedness is an operator inequality on the CFT stress tensor. A positive-energy Mack representation $(k, \bar{k}) = (4, 0)$, $\Delta = 7/2$, obeys ordinary unitarity and has double twist $-1/2$. Its generalized-free Wightman completion is conformally covariant and microcausal, and Wick powers make H_2 unbounded below. Translations in this generalized free theory lack a local stress tensor, placing it outside the CFT class used here. Lower boundedness of H_2 in a local CFT therefore requires a proof of (4.12) from stress-tensor locality and the all-spin ANEC constraints.

5 Supersymmetry in the large-spin contraction

Preserved supercharges are the zero-grade roots of the large-spin BPS generator. Their centralizer determines the contracted superalgebra and the scaling of protected indices.

5.1 Zero-grade centralizer

For Cartans (D, J_i, R_A) , define

$$K_\nu = D - \sum_{i \in \text{fast}} J_i - \nu^A R_A. \quad (5.1)$$

If an odd root F_α has weights $(d_\alpha, s_{\alpha i}, r_{\alpha A})$, set

$$\kappa_\alpha = d_\alpha - \sum_i s_{\alpha i} - \nu^A r_{\alpha A}. \quad (5.2)$$

Diagonalizing $\text{ad } K_\nu$ gives the centralizer

$$\boxed{\text{Cent}_{\mathfrak{g}}(K_\nu)_{\bar{1}} = \bigoplus_{\alpha: \kappa_\alpha=0} \mathfrak{g}_\alpha, \quad \text{Cent}_{\mathfrak{g}}(K_\nu)_{\bar{0}} = \mathfrak{h} \oplus \bigoplus_{\gamma: \kappa_\gamma=0} \mathfrak{g}_\gamma.} \quad (5.3)$$

The root grade contains R -weights and omits flavor weights because supercharges are flavor neutral.

On a large-charge ray labelled by Λ , take

$$\{F, F^\dagger\} = \Lambda^p b + O(\Lambda^{p-1}), \quad \widehat{F} = \Lambda^{-p/2} F. \quad (5.4)$$

The contracted bracket is $\{\widehat{F}, \widehat{F}^\dagger\} \rightarrow b$, with b central in the limiting representation. The rescaled transverse Heisenberg generators and these fermions form the super-Heisenberg extension. Zero-grade BPS roots with finite anticommutators retain their original normalization.

5.2 Three dimensions

For $\mathfrak{osp}(\mathcal{N}|4)$ and the uncharged generator $K_0 = D - J_z$, the odd zero-grade roots are

$$Q_+^I, \quad S_-^I, \quad I = 1, \dots, \mathcal{N}. \quad (5.5)$$

The even zero-grade generators are

$$D, \quad J_z, \quad P_+, \quad K_-, \quad R^{IJ}, \quad (5.6)$$

where R^{IJ} generate $\mathfrak{so}(\mathcal{N})$. In a conventional normalization their nontrivial odd brackets have the structure

$$\begin{aligned} \{Q_+^I, Q_+^J\} &\propto \delta^{IJ} P_+, & \{S_-^I, S_-^J\} &\propto \delta^{IJ} K_-, \\ \{Q_+^I, S_-^J\} &\propto \delta^{IJ} (D + J_z) + R^{IJ}. \end{aligned} \quad (5.7)$$

The brackets identify the preserved real-form algebra as $\mathfrak{osp}(\mathcal{N}|2) \oplus \mathfrak{u}(1)_{D-J_z}$: the $\mathfrak{sl}(2)$ factor is generated by $(P_+, K_-, D + J_z)$.

For an $\mathcal{N} = 2$ protected ray we take

$$q = Q_-, \quad \delta = \frac{1}{2} \{q, q^\dagger\} = D - J_z - R = \tau - R. \quad (5.8)$$

The zero-grade odd pair for this shifted generator is $(Q_-^{R=+1}, S_+^{R=-1})$, and its even centralizer is $\text{span}\{D, J_z, R, P_+, K_-\}$. The resulting complex algebra is $\mathfrak{sl}(2) \oplus \mathfrak{gl}(1|1)$, with real form understood and $\{q, q^\dagger\} = 2\delta$. The Regge-compatible BPS scaling is

$$J_z = \Lambda^2 j, \quad \tau = \Lambda h, \quad R = \Lambda h + O(1), \quad \delta = O(1). \quad (5.9)$$

The opposite spin component has an anticommutator of order Λ^2 , so division by Λ gives the finite fermionic oscillator in (5.4).

The protected trace is

$$\mathcal{I}_3(x, \mu_a) = \text{Tr}_{\mathcal{H}(S^2)} (-1)^F e^{-\beta\delta} x^{D+J_z} \prod_a \mu_a^{F_a}, \quad 0 < |x| < 1, \quad (5.10)$$

where F_a commute with q . The trace converges for $0 < |x| < 1$ and for flavor fugacities inside their convergence annulus. The large-spin edge uses

$$x = \exp(-\omega/\Lambda^2), \quad \text{Re } \omega > 0, \quad (5.11)$$

because $D + J_z \sim \Lambda^2$. Localization to $\delta = 0$ imposes $\tau = R$. Any separately resolved or saddle-selected ray with $J_z = \Lambda^2 j$ and $R = \Lambda h$ has $R \sim \tau \sim \sqrt{J}$. The trace (5.10) grades the combination $D + J_z = 2J_z + R$. Resolving J_z and R separately requires another commuting flavor charge. CFT data determine the cancellations and singular asymptotics as $x \rightarrow 1$.

5.3 Four dimensions

For the uncharged double-spin generator

$$K_0 = D - J_1 - J_2, \quad (5.12)$$

all odd roots have nonzero grade. Undotted and dotted supercharges give grades (in the KMP Cartan convention, respectively $-1/2$ or $3/2$, and $1/2$). The even centralizer is the grade-zero conformal algebra $\mathfrak{su}(1, 2) \oplus \mathfrak{u}(1)$, together with the R -symmetry algebra, which is the even root sum in (5.3). The zero-grade thermal centralizer is bosonic. Rescaling (5.4) can retain nonzero-grade odd generators in the contraction.

An R -shift restores a protected sector. In $\mathcal{N} = 4$ SYM we take a root with

$$(D, J_1, J_2; Q_1, Q_2, Q_3) = \left(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) \quad (5.13)$$

and

$$\delta = D - J_1 - J_2 - Q_1 - Q_2 - Q_3. \quad (5.14)$$

The selected $q, q^\dagger, \delta$ generate $\mathfrak{su}(1|1)$. Diagonalizing δ gives the zero-grade centralizer in (5.3), whose complexification is $\mathfrak{s}(\mathfrak{gl}(1|1) \oplus \mathfrak{gl}(3|3))/\mathbb{C}\mathbf{1}$. On q -cohomology, $\mathfrak{psu}(2, 1|3)$ acts faithfully, up to real-form convention. Projectivization gives its even part $\mathfrak{su}(2, 1) \oplus \mathfrak{su}(3)$. A generic state on the BPS charge ray is annihilated by the selected complex supercharge and its radial adjoint.

We use the modified raw-charge trace

$$\mathcal{Z}_4(\Delta_I, \omega_i) = \text{Tr}_{\delta=0} \exp \left[- \sum_{I=1}^3 \Delta_I Q_I - \sum_{i=1}^2 \omega_i J_i \right], \quad \sum_I \Delta_I - \omega_1 - \omega_2 = 2\pi i \pmod{4\pi i}. \quad (5.15)$$

On this sheet the chemical exponential supplies the fermion sign, so a second factor of $(-1)^F$ defines a different trace. In the large- N Cardy chamber with $|\omega_i| \ll 1$, complex $O(1)$ Δ_I , and the saddle selected by analytic continuation from its convergence domain,

$$\log \mathcal{Z}_4 \sim \frac{N^2 \Delta_1 \Delta_2 \Delta_3}{2\omega_1 \omega_2}. \quad (5.16)$$

Refs. [11, 12] derive (5.16) in this chamber. Analytic continuation to another sheet can exchange the dominant saddle.

Differentiating (5.16) gives, in magnitude,

$$Q_I \sim \frac{N^2}{\omega_1 \omega_2}, \quad J_1 \sim \frac{N^2}{\omega_1^2 \omega_2}, \quad J_2 \sim \frac{N^2}{\omega_1 \omega_2^2}. \quad (5.17)$$

On $\delta = 0$, $\tau = D - J_1 - J_2 = \sum_I Q_I$. Substitution in (5.17) gives

$$\boxed{\tau \sim (J_1 J_2)^{1/3}}. \quad (5.18)$$

When N is held fixed while $\omega_i \rightarrow 0$ at fixed ω_1/ω_2 , $\tau/(J_1 J_2)^{1/3} \sim N^{2/3}$. A simultaneous large- N scaling must specify that combination. Anomaly coefficients set the leading coefficient, with saddle exchange and cancellations controlled by the analytic chamber and the CFT spectrum.

6 Kerr limits with pp-wave boundaries

Ultraspinning limits of Kerr–AdS produce exact Einstein metrics whose conformal boundaries are the KMP pp waves. Fefferman–Graham data determine the holographic stress tensors, and the outer horizons determine the thermal quotients.

6.1 Double-spin Einstein solution

The equal-two-rotation ultraspinning limit of neutral Kerr–AdS₅ constructed by Maldacena, Martelli and Tachikawa (MMT) [10] has boundary metric

$$g_{(0)} = -dx^+ (dx^- - 2\rho^2 d\hat{\psi}) + d\rho^2 + \rho^2 d\hat{\psi}^2. \quad (6.1)$$

The coordinate map to the KMP frame is

$$x^+ = t, \quad u = \rho \cos \hat{\psi}, \quad y = -\rho \sin \hat{\psi}, \quad x^- = t + 2(v + uy). \quad (6.2)$$

Substitution gives $dx^- - 2\rho^2 d\hat{\psi} = dt + 2dv + 4u dy$ and maps (6.1) to (2.5).

In these coordinates the full solution is

$$\boxed{\frac{ds_5^2}{L^2} = \frac{dR^2}{R^2 F(R)} + R^2 g_{\text{KMP}} - dt^2 + \frac{m\lambda^6}{2R^2} [2dv + 4u dy + (1 + \lambda^{-2})dt]^2}, \quad (6.3)$$

$$F(R) = 1 - \frac{2m\lambda^4}{R^4} + \frac{2m\lambda^6}{R^6}, \quad R_{\mu\nu} = -4L^{-2}g_{\mu\nu}. \quad (6.4)$$

The metric and Einstein equation follow by smooth scaling from Kerr–AdS₅. Consistency of $F(R)$ requires the λ^6 term. The λ^4 printed below eq. (3.24) of ref. [10] is a typo.

Set $\Theta = 2dv + 4u dy + (1 + \lambda^{-2})dt$. The Fefferman–Graham expansion gives

$$g_{(2)} = -dt^2, \quad g_{(4)} = \frac{m\lambda^6}{2}\Theta^2 + \frac{m\lambda^4}{2}g_{\text{KMP}}, \quad \text{tr}_{g_{(0)}}g_{(4)} = 0. \quad (6.5)$$

Proposition 6.1 (Horizon regularity). *Let $x = R_H^2/\lambda^2$. The largest-root branch is*

$$m = \frac{x^3}{2(x-1)}, \quad T = \frac{2x-3}{2\pi\sqrt{x-1}}, \quad x > \frac{3}{2}. \quad (6.6)$$

The simple root has a horizon Killing vector that generates the smooth Euclidean thermal identification.

Proof. The equation $F(R_H) = 0$ gives the first expression. The function $m(x)$ has its minimum at $x = 3/2$. Choosing $x > 3/2$ selects the outer simple root with $F'(R_H) \neq 0$. Its surface gravity gives the displayed $T > 0$. The null generator in MMT coordinates is

$$\chi = \partial_{x^+} + \frac{x-2}{x\lambda^2} \partial_{x^-}. \quad (6.7)$$

The complex Euclidean quotient generated by $\beta\chi$, $\beta = T^{-1}$, removes the polar-cone singularity. Smooth scaling of the Kerr outer horizon maps its Euclidean identification to this local quotient. $\square$

We take $x^- \in \mathbb{R}$, so the horizon has infinite area and finite density. Compactifying x^- would add global data not present in the local pp-wave limit. In the bulk, $g_{--} = m\lambda^6/(2R^2) > 0$, so the x^- direction remains spacelike through the outer horizon.

MMT’s compact-coordinate normalization gives

$$\frac{N}{V_2} = \frac{L^3}{4G_5} r^- m\lambda^6, \quad \frac{H_{\text{MMT}}}{V_2} = \frac{L^3}{4G_5} r^- \frac{m\lambda^4}{2}. \quad (6.8)$$

The coordinate map (6.2) changes the time Killing field:

$$\partial_t^{\text{KMP}} = \partial_{x^+} + \partial_{x^-}, \quad H_{\text{KMP}} = H_{\text{MMT}} + \frac{N}{r^-}. \quad (6.9)$$

6.2 Unequal-spin Regge scaling

For arbitrary a_1, a_2 , the Kerr-AdS₅ charges determine the unequal-spin approach to the Regge ray. We set

$$A_i = 1 - a_i, \quad K = \frac{\pi m}{16G_5}. \quad (6.10)$$

The standard charges of ref. [9] obey the exact identity

$$\tau = E - J_1 - J_2 = \frac{\pi m(4 - A_1 A_2)}{4G_5 A_1 A_2 (2 - A_1)^2 (2 - A_2)^2}. \quad (6.11)$$

As $A_i \rightarrow 0^+$,

$$\tau \sim \frac{K}{A_1 A_2}, \quad J_1 \sim \frac{K}{A_1^2 A_2}, \quad J_2 \sim \frac{K}{A_1 A_2^2}, \quad \frac{\tau}{(J_1 J_2)^{1/3}} \longrightarrow K^{1/3}. \quad (6.12)$$

The physical regulators are

$$\epsilon_i = 1 - \Omega_i = A_i \frac{r_+^2 - a_i}{r_+^2 + a_i^2} = c A_i + O(A_i^2), \quad c = \frac{r_+^2 - 1}{r_+^2 + 1}. \quad (6.13)$$

At leading order,

$$\tau = \frac{K c^2}{\epsilon_1 \epsilon_2}, \quad J_1 = \frac{K c^3}{\epsilon_1^2 \epsilon_2}, \quad J_2 = \frac{K c^3}{\epsilon_1 \epsilon_2^2}, \quad \frac{J_1}{J_2} = \frac{\epsilon_2}{\epsilon_1}. \quad (6.14)$$

The regulator Jacobian gives $\text{Vol}_{\text{pw}} = \pi^2 / (2\epsilon_1 \epsilon_2)$. On the outer branch,

$$T = \frac{2r_+^2 - 1}{2\pi r_+}, \quad \frac{S}{\text{Vol}_{\text{pw}}} = \frac{(r_+^2 - 1)^2}{4G_5 r_+}, \quad \frac{I_E}{\beta \text{Vol}_{\text{pw}}} = -\frac{(r_+^2 - 1)^3}{16\pi G_5 r_+^2}, \quad (6.15)$$

reproducing (3.3). In this limit the boundary angular velocities approach light speed at fixed r_+ , and the azimuths decompactify. Super-entropic limits add a compactification after this scaling. Kerr/CFT instead introduces near-horizon coordinates, which are a different scaling. A local Einstein metric with unequal spins and a pp-wave boundary remains to be constructed.

6.3 BPS charge relation

Supersymmetric rotating three-charge AdS₅ black holes obey the $U(1)^3$ gauged-supergravity constraint [11, 13]

$$Q_1 Q_2 Q_3 + \frac{N^2}{2} J_1 J_2 - \left(\sum_I Q_I + \frac{N^2}{2} \right) \left(\sum_{I < J} Q_I Q_J - \frac{N^2}{2} (J_1 + J_2) \right) = 0. \quad (6.16)$$

On the Cardy/pp ray, its leading terms give

$$\left(\sum_I Q_I \right) \left(\sum_{I < J} Q_I Q_J \right) - Q_1 Q_2 Q_3 \sim \frac{N^2}{2} J_1 J_2. \quad (6.17)$$

For equal $Q_I = Q$ and $J_i = J$,

$$\left(\frac{J}{N^2}\right)^2 \sim 16 \left(\frac{Q}{N^2}\right)^3. \quad (6.18)$$

For equal charges, $\tau = 3Q$, so (6.18) has the scaling $\tau \sim (J_1 J_2)^{1/3}$ found in (5.18). Equation (6.3) has zero gauge fields and equal spins. The charge relation supplies a necessary condition for a charged arbitrary-ratio solution.

6.4 One-spin AdS₄ solution

We start from the Kerr–AdS₄ Boyer–Lindquist metric [9] and take

$$t = x^+, \quad \phi = -\frac{x^-}{2\mathcal{R}^2} + (1-a)x^+, \quad 1-a = \frac{1}{2\lambda\mathcal{R}^2}, \quad \mathcal{R} \longrightarrow \infty, \quad \lambda > 0. \quad (6.19)$$

Define

$$\Sigma = r^2 + \sin^2 \theta, \quad \Delta = (1+r^2)^2 - 2mr. \quad (6.20)$$

Taking the limit term by term gives

$$\boxed{\begin{aligned} \frac{ds_4^2}{L^2} = & \frac{\Sigma}{\Delta} dr^2 + \frac{\Sigma}{\cos^2 \theta} d\theta^2 - (1+r^2 \sin^2 \theta)(dx^+)^2 \\ & - \lambda(1+r^2) \cos^2 \theta dx^+ dx^- \\ & + \frac{mr}{2\Sigma} [-2dx^+ + (dx^+ - \lambda dx^-) \cos^2 \theta]^2. \end{aligned}} \quad (6.21)$$

The prelimit metrics and their first two derivatives converge smoothly on compact subsets of the exterior. Because (6.21) is a local noncompact limit of exact Einstein metrics, it obeys $R_{\mu\nu} = -3L^{-2}g_{\mu\nu}$. The scaling decompactifies x^- and turns the Kerr horizon into a noncompact surface.

On $-\pi/2 < \theta < \pi/2$, we introduce the signed coordinates

$$Y = \sqrt{\lambda(1+r^2)} \cos \theta, \quad \rho = \frac{r \sin \theta}{Y}. \quad (6.22)$$

In terms of $A = 1 + \lambda\rho^2$ and

$$W = (1 + 2\lambda\rho^2)dx^+ + \lambda dx^-, \quad (6.23)$$

a Fefferman–Graham defining radius R begins as

$$Y = R + \frac{m\lambda^{3/2}}{3A^{3/2}R^2} + O(R^{-4}), \quad (6.24)$$

and the radial cross term is removed by

$$\rho_{\text{raw}} = \rho + \frac{m\rho\lambda^{5/2}}{5A^{5/2}R^5} + O(R^{-7}). \quad (6.25)$$

The expansion

$$\frac{ds_4^2}{L^2} = \frac{dR^2}{R^2} + R^2 [g_{(0)} + R^{-2}g_{(2)} + R^{-3}g_{(3)} + O(R^{-4})] \quad (6.26)$$

has

$$\begin{aligned}
g_{(0)} &= -dx^+ dx^- - \rho^2 (dx^+)^2 + d\rho^2, \\
g_{(2)} &= -(dx^+)^2, \\
g_{(3)} &= \frac{2m\lambda^{3/2}}{3A^{3/2}} g_{(0)} + \frac{m\sqrt{\lambda}}{2A^{5/2}} W^2.
\end{aligned} \tag{6.27}$$

$\text{tr}_{g_{(0)}} g_{(3)} = 0$ and $\nabla_{(0)}^a g_{(3)ab} = 0$, as required for the holographic stress tensor. Setting $x^+ = t$ and $x = t - x^-$ turns $g_{(0)}$ into the physical KMP metric (2.1).

The largest outer root r_H satisfies

$$m = \frac{(1 + r_H^2)^2}{2r_H}, \quad T = \frac{3r_H^2 - 1}{4\pi r_H}. \tag{6.28}$$

The root is simple for $r_H > 1/\sqrt{3}$, including the stable Hawking–Page branch $r_H \geq 1$. Its horizon generator is

$$\chi = \partial_{x^+} + \frac{r_H^2 - 1}{\lambda(r_H^2 + 1)} \partial_{x^-}. \tag{6.29}$$

The complex Euclidean quotient by $\beta\chi$, $\beta = T^{-1}$, is the analytic continuation of the smooth Kerr rotating saddle and removes the local conical singularity. Global reality fails for the Euclidean continuation of the Lorentzian pp wave. The original azimuthal period sends the period of x^- to infinity in (6.19). We retain $x^- \in \mathbb{R}$, giving infinite horizon area with finite density. The limits $\theta \rightarrow \pm\pi/2$ are infinite-distance cusps. A finite x^- quotient would add DLCQ-like global data.

The prelimit charges obey

$$E = \frac{m}{G_4(1 - a^2)^2}, \quad J = \frac{ma}{G_4(1 - a^2)^2}, \quad \tau \sim \frac{m}{4G_4(1 - a)}, \quad J \sim \frac{m}{4G_4(1 - a)^2}, \tag{6.30}$$

and

$$\epsilon = 1 - \Omega = \frac{r_H^2 - 1}{r_H^2 + 1} (1 - a) + O((1 - a)^2), \quad \frac{\tau}{\sqrt{J}} \longrightarrow \frac{1}{2} \sqrt{m/G_4}. \tag{6.31}$$

We define entropy densities against x^- and the regulated Kerr volume separately. Per unit x^- ,

$$\frac{dS}{dx^-} = \frac{L^2 \lambda (1 + r_H^2)}{4G_4}. \tag{6.32}$$

For the regulated Kerr volume, the area law gives

$$\frac{S}{2\pi/\epsilon} = \frac{L^2 (r_H^2 - 1)}{4G_4}, \tag{6.33}$$

which equals $d\mathcal{F}_3/dT$ in the $L = 1$ conventions of section 3. Equation (6.33) uses the effective volume $\text{Vol}_{\text{pw}} = 2\pi/\epsilon$. Its positive-regulator branch has $r_H > 1$ and a continuous endpoint at $r_H = 1$.

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